

Δημιουργία RAG αρχιτεκτονικής για ερωτήσεις φυσικής γλώσσας σε ειδησεογραφικά άρθρα

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Αρχικά αυτό που έκανα είναι να ανοίξω το αρχείο CNN_Articels_clean.csv, και να κρατήσω μόνο τα πεδία headline ως meta_field και το Article text που θεώρησα ότι είναι και το πιο σημαντικό διότι αυτό εμπεριέχει την μεγαλύτερη πληροφορία. Δεν χρησιμοποίησα παραπάνω meta fields διότι δεν γνωρίζω πως ακριβώς δούλευαν. Η όλη διαδικασία γίνεται στο παρακάτω κομμάτι κώδικα

```
documents = []
for _, row in df.iterrows():
    text = str(row[text_field])
    metadata = {field: str(row[field]) for field in meta_fields}
    documents.append(Document(page_content=text, metadata=metadata))
```

(α) Διαχωρισμός του κειμένου σε μικρότερα τμήματα(chunk).

Για τον σκοπό αυτό χρησιμοποίησα το RecursiveCharacterTextSplitter που παρέχει το LangChain

```
text_splitter = RecursiveCharacterTextSplitter(
    chunk_size=500,
    chunk_overlap=100,
    separators=["\n\n", "\n", ".", "!", "?", ",", " ", " "]
)
```

Όπως βλέπετε έχω φτιάξει ένα προφίλ για τα chunks μου όπου θα έχουν μέγεθος 500 , θα γίνεται overlap τα 100 από αυτά , έτσι ώστε κάποιες προτάσεις να ολοκληρώνονται και να μην αποκόπτονται απρόσμενα
Π.χ.

Χωρίς overlap :

Chunk 1: "Albert Einstein was born in Ulm, Germany in 1879. He developed the theory"

Chunk 2: "of relativity, one of the two pillars of modern physics."

Με overlap:

Chunk 1: "Albert Einstein was born in Ulm, Germany in 1879. He developed the theory"

Chunk 2: "He developed the theory of relativity, one of the two pillars of modern physics."

Κάποια σημεία είναι κοινά αλλά δίνει την ευκαιρία να ολοκληρωθούν οι προτάσεις.

Εφόσον έχω πλέον δημιουργήσει το προφίλ τους απλά το εφαρμόζω πάνω στα κείμενα των άρθρων

```
chunked_documents = text_splitter.split_documents(documents)
```

(β) Δημιουργία διανυσμάτων για κάθε τμήμα(chunk):

Για την δημιουργία των επιμέρους διανυσμάτων (embeddings) χρησιμοποίησα το παρακάτω model:

```
model = SentenceTransformer('all-MiniLM-L6-v2')  
Ακολουθεί ο παρακάτω κώδικας για την δημιουργία τους
```

```
embedding_file = "embeddings.npy"  
if os.path.exists(embedding_file):  
  
    embeddings = np.load(embedding_file)  
    print("Embeddings loaded from 'embeddings.npy'.")  
else:  
    print("Generating embeddings for each chunk...")  
    embeddings = []  
    for i, doc in enumerate(chunked_documents):  
        embedding = model.encode(doc.page_content)  
        embeddings.append(embedding)  
        if i % 100 == 0:  
            print(f"Processed {i+1}/{len(chunked_documents)} chunks...")
```

Στον παραπάνω κώδικα , τα embeddings αυτά δημιουργούνται αν δεν υπάρχουν ήδη για να μην δημιουργούνται κάθε φορά από την αρχή.

Παίρνουμε ένα-ένα τα chunks και δημιουργούμε embeddings μόνο για το page_content , το οποίο αναφέρεται μόνο στα πεδία των άρθρων που είναι "Article text" , έχω βάλει και ένα meta_field το "Headline" , αλλά δεν το χρησιμοποίησα για την δημιουργία embeddings,καθώς θα ήταν αρκετά περίπλοκη η διαχείριση του μετά. Τα embeddings που δημιουργούνται από το μοντέλο all-MiniLM-L6-v2 τα αποθηκεύω στο αρχείο embeddings.npy και αποτελεί "την βάση δεδομένων μου"

(γ) Ανάκτηση relevant chunks.

Στην δικιά μου περίπτωση για την ομοιότητα του query και των embeddings απο βάση δεδομένων χρησιμοποίησα τα παρακάτω

```
dimension = embeddings.shape[1]
index = faiss.IndexFlatL2(dimension)
index.add(embeddings)
```

Το **FAISS** (Facebook AI Similarity Search) , είναι και αυτό της meta , για vector searching , IndexFlatL2 το συγκεκριμένο συγκρίνει τα 2 embeddings με ευκλείδεια αντι για cosine similarity που αναφέρεται στην εκφώνηση της άσκησης , είναι αρκετά γρήγορο και αποτελεσματικό για μικρά δεδομένα.

Βέβαια για να γίνει αυτό μετατρέπω και το query σε embedding στον παρακάτω κώδικα πάλι μέσω του model all-MiniLM-L6-v2

```
query = input("Enter your query: ")
query_embedding = model.encode(query).astype('float32')
```

Εφόσον δημιουργηθεί το embedding χρησιμοποιώ το index που όρισα παραπάνω ως εργαλείο σύγκρισης για να συγκρίνω τα chunks με το query

```
D, I = index.search(np.array([query_embedding]), k=5)
```

Σε αυτήν την περίπτωση απλά ζητάω τα 5 πιο relevant chunks , μπορείτε να αλλάξετε το k αν θέλετε παραπάνω chunks

Έπειτα τυπώνω ένα μέρος των chunks , όπως όρισα παραπάνω το chunk size είναι 500 αλλά εγώ τυπώνω μόνο τα πρώτα 200 στοιχεία

```
for i in range(5):
    print(f"\n--- Result {i+1} ---")
    print("Document Metadata:", chunked_documents[I[0][i]].metadata)
    print("Content:", chunked_documents[I[0][i]].page_content[:200], "...")
```

Έχω αλλάξει το κώδικα έτσι ώστε τα k top documents να ορίζονται απο τον χρήση μέσω terminal

Τα k-top documents λοιπον τα συνενώνω και τα ορίζω ως context

```
retrieved_chunks = [chunked_documents[i].page_content for i in I[0]]
context = "\n\n".join(retrieved_chunks)
```

Σχετικά με το prompt , έχω ορίσει διάφορες προτάσεις και επιλέγεται τυχαία μια από αυτές χρησιμοποιώντας το query του χρήστη και το context που ανακτήθηκε από την "βάση δεδομένων" , μπορούμε να προσθέσουμε και άλλες προτάσεις αν θέλουμε

```

prompt_templates = [
    """
    You are an AI assistant. Answer the following question: "{query}",
    based only on the following context:\n\n{context}
    """,

    """
    Based on the following context, provide a response to the query: "{query}".
    Context:\n\n{context}
    """,

    """
    You have the following information. Answer the question: "{query}" using only
    the context below:\n\n{context}
    """,

    """
    Please respond to the following query: "{query}" by referring to the given
    context:\n\n{context}
    """,

    """
    Given the context below, provide the best possible response to the query:
    "{query}".\n\n{context}
    """,

    """
    Answer the question: "{query}", using only the context provided below:
    \n\n{context}
    """]

```

```

selected_template = random.choice(prompt_templates)

```

```

before_query, after_query = selected_template.split('{query}')
between_query_context, _ = after_query.split('{context}')

```

```

formatted_prompt = selected_template.format(query=query, context=context)

```

Ο λόγος που το έκανα αυτό είναι για να έχουμε μια πληθώρα prompt για το llm που θα βοηθήσει μετέπειτα στο evaluation και όχι μια καθολική πρόταση.

(δ) Επαύξηση του LLM prompt

Τέλος στο υπόλοιπο κομμάτι κώδικα χρησιμοποιώ το llm ollama , 2 φορές κάνοντας prompt μια φορά μόνο με το query του χρήση και αμέσως μετά συνδυασμός του query μαζί με το context που ανακτήθηκε από τα top k relevant chunks για σύγκριση

```
def query_ollama(prompt, model="llama3.2:1b"):
    response = requests.post(
        "http://localhost:11434/api/generate",
        json={
            "model": model,
            "prompt": prompt,
            "stream": False
        }
    )
    if response.status_code == 200:
        return response.json()["response"]
    else:
        print("Error:", response.status_code, response.text)
        return None

print("\nQuerying Ollama...without context: ",query)
response = query_ollama(query)
print("\n🗨️ LLM Response:\n", response)
# Query Ollama with formatted prompt
print("\nQuerying Ollama...")
print(before_query.strip(), "{query}" ,between_query_context.strip() , "{context}")
response = query_ollama(formatted_prompt)
print("\n🗨️ LLM Response:\n", response)
```

(ε) Evaluation του συστήματος

1.

Querying Ollama...without context: Why CNN is launching a new series on skin whitening?

Top 5 relevant chunks:

--- Result 1 ---

Document Metadata: {'Headline': 'Skin Whitening: The global market, the risks and who makes and sells the products - CNN'}

Content: This story is part of 'White lies', a series by CNN's As Equals investigating skin whitening practices worldwide to expose the underlying drivers of colorism, the industry that profits from it and the ...

--- Result 2 ---

Document Metadata: {'Headline': 'Why CNN is launching a new series on skin whitening - CNN'}

Content: This story is part of 'White lies', a series by CNN's As Equals investigating skin whitening practices worldwide to expose the underlying drivers of colorism, the industry that profits from it and the ...

--- Result 3 ---

Document Metadata: {'Headline': 'Skin Whitening: The global market, the risks and who makes and sells the products - CNN'}

Content: ."eBay removed the listings highlighted by CNN that were selling some of these products.Read: Why CNN is launching a new series on skin whitening ...

--- Result 4 ---

Document Metadata: {'Headline': 'Why CNN is launching a new series on skin whitening - CNN'}

Content: . But skin lightening, whitening or bleaching is not just a Global South problem -- the US makes up a third of the 2020 market. Read MoreThis harmful beauty standard affects billions of people around ...

--- Result 5 ---

Document Metadata: {'Headline': 'Skin Whitening: The global market, the risks and who makes and sells the products - CNN'}

Content: .Shivanya, Faiza, Chandni and Noor did not respond to CNN's requests for comment. Social media companies were asked whether they were aware that skin whitening creams, including ones containing toxic ...

 LLM Response:

I'm not aware of any recent information that CNN is launching a new series specifically on skin whitening. However, I can try to provide some context and possible reasons behind such a move.

It's worth noting that there has been a growing trend in media coverage surrounding skin whitening and beauty products, particularly among certain demographics. Some critics have raised concerns about the industry's impact on body image, racial equity, and consumer safety.

CNN, as a news organization, may be addressing this issue by launching a new series or report on the topic of skin whitening. Here are some possible reasons:

1. **Racial equity**: CNN has been working to address issues of racial inequality in media coverage over the years. Launching a series on skin whitening could be part of their effort to provide accurate and nuanced reporting on this complex issue.
2. **Consumer safety**: The beauty industry is known for its vast range of products, many of which are sold without clear labeling or regulations. CNN may be exploring ways to educate viewers about the potential risks associated with certain skin whitening products.
3. **Pop culture commentary**: Skin whitening has become a significant aspect of popular culture, particularly in the realm of beauty and fashion. CNN might use this series as an opportunity to comment on the cultural significance of skin whitening and its impact on societal norms.

It's essential to note that without more specific information or credible sources, it's difficult to confirm whether CNN is indeed launching a new series on skin whitening. If you have any further questions or concerns, feel free to ask.

Querying Ollama...

You have the following information. Answer the question: " {query} " using only the context below: {context}



LLM Response:

The CNN series "White lies: Exposing the dangers of skin whitening" is launching because it aims to investigate how skin whitening practices are used worldwide, particularly in the Global South, as part of a larger exploration into colorism and the industry that profits from it.

2.

Querying Ollama...without context: why greece has built a wall on border with turkey ?

Top 5 relevant chunks:

--- Result 1 ---

Document Metadata: {'Headline': 'Greece finishes wall on border with Turkey, amid fears of Afghan migrant crisis - CNN'}

Content: (CNN)Greece has finished building a 40-kilometer (25-mile) wall along its border with Turkey, amid concerns in parts of Europe that the Taliban's takeover of Afghanistan could cause an influx of peopl ...

--- Result 2 ---

Document Metadata: {'Headline': 'Greece finishes wall on border with Turkey, amid fears of Afghan migrant crisis - CNN'}

Content: .The country was at the center of Europe's migrant crisis in the mid-2010s, when millions of refugees from Syria, Afghanistan and Iraq traveled to the continent. Since then Greece has taken a hardline ...

--- Result 3 ---

Document Metadata: {'Headline': "Europe's disunity and lack of trust imperil the continent's future - CNN"}

Content: . A group of migrants from Syria walk towards the border with Hungary, near the northern Serbian village of Martonos, near Kanjiza, on June 25, 2015.In 2016, the EU gave Turkey -- a regional partner - ...

--- Result 4 ---

Document Metadata: {'Headline': 'Greece finishes wall on border with Turkey, amid fears of Afghan migrant crisis - CNN'}

Content: ."Even before the Taliban took Afghanistan, the pandemic had worsened the migrant crisis. A deal was struck between Turkey and the European Union in 2016 to halt migrants traveling from the Middle Eas ...

--- Result 5 ---

Document Metadata: {'Headline': ' As Joe Biden walks America away from the world stage Europe is left exposed - CNN'}

Content: . However, Brussels tried this back in 2015 when it gave Turkish President Recep Tayyip Erdogan's government money to stem the flow of migrants into the bloc. "They've learned all the wrong lessons f ...

... LLM Response:

You're referring to the recent development in Greece-Turkey relations. The construction of a fence or barrier along the eastern Aegean coast of Turkey, which is part of Greece, has been a topic of concern for many countries and organizations due to potential security implications.

The main reason for this action was a dispute over the island of Kastellorizo, which is located in the northern part of the Turkish-occupied area known as the Northern East Aegean. In 2016, Greece launched an operation to reclaim the islands from Turkey, but the operation was ultimately unsuccessful.

As a result, the international community and local groups raised concerns about potential human rights abuses, including extrajudicial killings, torture, and arbitrary detention of Turkish nationals and Greek refugees on Kastellorizo. The UN Human Rights Council also expressed concerns about the situation.

In response to these concerns, Greece decided to build a physical barrier along its coastline to prevent migrants and asylum seekers from attempting to reach Kastellorizo. The barrier is not intended as a punishment or an attempt to restrict Turkey's sovereignty but rather to protect Greek citizens and prevent harm to them.

Additionally, the construction of this border was also driven by concerns about the movement of military equipment and supplies into the disputed zone. Some reports suggest that Turkish forces have been transporting troops and military vehicles through the area in recent months.

It's worth noting that the situation remains complex, and there are ongoing disputes between Greece and Turkey over issues such as border demarcation, maritime boundaries, and access to ports and natural resources.

Querying Ollama...

Answer the question: " {query} ", using only the context provided below: {context}



LLM Response:

Based on the provided context, Greece has built a wall along its border with Turkey because it aims to reduce the flow of migrants across its borders and is concerned about an influx of people seeking asylum due to concerns in Europe that the Taliban's takeover of Afghanistan could cause an increase in migrant flows.

3.

Querying Ollama...without context: which are the fastest growing economies ?

Top 5 relevant chunks:

--- Result 1 ---

Document Metadata: {'Headline': "Opinion: Media coverage of Ukraine shows it's time to rethink what we know about Africa – CNN"}

Content: . It's the reason why the Twittersphere erupted recently when journalist Alan Macleod put together his greatest hits of racist coverage of the Ukraine invasion -- thank you, Alan! Contrary to what ne ...

--- Result 2 ---

Document Metadata: {'Headline': "India's young investors prefer crypto to gold and 'boring' stocks – CNN"}

Content: . Entrepreneurs in the industry told CNN Business that the country has the potential to become a crypto superpower, since it is one of the hottest internet markets in the world, with 750 million users ...

--- Result 3 ---

Document Metadata: {'Headline': "Opinion: Media coverage of Ukraine shows it's time to rethink what we know about Africa – CNN"}

Content: . Given all this digital activity, it is hardly surprising that Africa is one of the fastest-growing markets in terms of venture capital activities and financial technology. Africa is also regarded as ...

--- Result 4 ---

Document Metadata: {'Headline': 'Economic growth in 2022: China had a surprisingly good start, but it may not last – CNN'}

Content: . That was well above the estimated 3% increase in a Reuters poll of economists. Industrial production jumped 7.5% during the same period, surpassing the forecast of 3.9%. And investment in fixed asset ...

--- Result 5 ---

Document Metadata: {'Headline': "Opinion: Media coverage of Ukraine shows it's time to rethink what we know about Africa – CNN"}

Content: . But with an average GDP growth predicted of 6.7 between 2021–2025, it is in the top five of the fastest-growing economies alongside India. Rwanda has been able to attract foreign direct investment a ...

... LLM Response:

According to various sources, including the International Monetary Fund (IMF) and the World Bank, here are some of the fastest-growing economies in the world:

1. **Nigeria**: With a GDP growth rate of around 7-8% per annum, Nigeria has been one of the fastest-growing economies in Africa over the past decade.
2. **Bangladesh**: Bangladesh has experienced rapid economic growth, driven by its large and growing population, as well as investments in infrastructure and industries such as textiles and electronics.
3. **Sri Lanka**: Sri Lanka's economy has been experiencing steady growth, thanks to its strategic location, tourism industry, and favorable business environment.
4. **Vietnam**: Vietnam has emerged as a major economic power in Southeast Asia, driven by its rapid industrialization and urbanization processes.
5. **Poland**: Poland has made significant progress in recent years, with a GDP growth rate of around 3-4% per annum. Its economy is driven by its automotive industry, manufacturing sector, and services sector.
6. **Turkey**: Turkey's economy has experienced steady growth, thanks to its strategic location, tourism industry, and favorable business environment.
7. **Indonesia**: Indonesia's economy has been growing steadily, with a GDP growth rate of around 5-6% per annum. Its economy is driven by its manufacturing sector, tourism industry, and natural resources.
8. **India**: India's economy has emerged as one of the largest in the world, with a GDP growth rate of around 7-8% per annum. Its economy is driven by its large and growing population, as well as investments in infrastructure and industries such as IT and pharmaceuticals.
9. **Mexico**: Mexico's economy has experienced steady growth, thanks to its strategic location, tourism industry, and favorable business environment.
10. **Brazil**: Brazil's economy has been experiencing slow but steady growth, driven by its large and growing population, as well as investments in infrastructure and industries such as oil and gas.

Please note that these figures are subject to change and may vary depending on various factors such as economic indicators, global events, and policy changes.

Querying Ollama...

You have the following information. Answer the question: " {query} " using only the context below: {context}



LLM Response:

The fastest-growing economies mentioned in the article are:

1. India
2. Vietnam

These two economies are ranked second and third in Africa's fastest growth rate behind Rwanda.

4.

Querying Ollama...without context: were there any data breach involving big companies ?

Top 5 relevant chunks:

--- Result 1 ---

Document Metadata: {'Headline': "Epik is a refuge for the deplatformed far right. Here's why its CEO insists on doing it - CNN"}

Content: . Lightning struck this fall, when so-called hacktivists illegally cracked open the company's databases and made them public, triggering a feeding frenzy for internet researchers who have been sifting ...

--- Result 2 ---

Document Metadata: {'Headline': 'UK Phone Hacking Scandal Fast Facts - CNN'}

Content: (CNN)Here's a look at the phone hacking scandal in the United Kingdom. Journalists at British newspapers are accused of making payments to police and hacking into the phones of celebrities, law makers ...

--- Result 3 ---

Document Metadata: {'Headline': 'Police arrest eight after celebrities hit by SIM-swapping attacks - CNN'}

Content: .One man lost his life savings in a SIM hack. Here's how you can try to protect yourselfThis allowed them to steal money, bitcoin and personal information, as well as hack their victims' social media ...

--- Result 4 ---

Document Metadata: {'Headline': "First Spanish-made 'narco sub' found in Mediterranean warehouse - CNN"}

Content: .Authorities investigated the network -- members of which had attempted to cover their tracks by setting up legitimate businesses -- for more than a year, according to a statement from Spain's national ...

--- Result 5 ---

Document Metadata: {'Headline': 'Shenzhen lockdown: Foxconn halts operations as Covid hits tech hub - CNN'}

Content: . China's Xi'an lockdown hits some of the world's largest chipmakersShanghai, the country's largest business center, has imposed stringent measures after a spike in Covid cases, closing schools and ...

 LLM Response:

Yes, there have been several high-profile data breaches involving big companies in recent years. Here are some notable examples:

1. ****Equifax Breach (2017)****: Equifax, one of the largest credit reporting agencies in the US, suffered a massive data breach that exposed sensitive information for over 147 million people. The breach was revealed to be a result of a vulnerability in the company's software.
2. ****Yahoo! Breach (2014-2016)****: Yahoo! suffered a series of major data breaches between 2013 and 2016, exposing sensitive information such as passwords, email addresses, and credit card numbers for over 3 billion users.
3. ****Target Breach (2013)****: Target, one of the largest retailers in the US, suffered a massive data breach that exposed sensitive information for over 40 million customers. The breach was attributed to a combination of physical and cyber attacks.
4. ****Walmart Breach (2015-2016)****: Walmart suffered a data breach that exposed sensitive information for over 1.1 million employees and 41,000 customers.
5. ****Home Depot Breach (2014)****: Home Depot suffered a data breach that exposed sensitive information for over 56,000 customers.
6. ****H&M Breach (2018)****: H&M, the fast-fashion retailer, suffered a data breach that exposed sensitive information for over 45 million customers.
7. ****Nike Breach (2017)****: Nike suffered a data breach that exposed sensitive information for over 500,000 employees and 750,000 customers.
8. ****Uber Breach (2016-2017)****: Uber suffered two major data breaches in 2016 and 2017, exposing sensitive information for over 56 million users.
9. ****Citi Breach (2017)****: Citi, one of the largest banks in the world, suffered a data breach that exposed sensitive information for over 130 million customers.
10. ****HSBC Breach (2015-2016)****: HSBC, one of the largest banks in the world, suffered a data breach that exposed sensitive information for over 1.3 billion customer accounts.

These are just a few examples of high-profile data breaches involving big companies. There have been many other breaches as well, resulting in significant financial and reputational harm to affected organizations.

Querying Ollama...

You are an AI assistant. Answer the following question: " {query} ",
based only on the following context: {context}



LLM Response:

Based on the provided context, here are some notable data breach incidents involving big companies:

1. News Corp.'s News of the World: A phone hacking scandal involving journalists at British newspapers who allegedly hacked into phones of celebrities, politicians, and others.
2. SIM hack victims: Several high-profile individuals, including a man losing his life savings in a SIM hack, have been reported to have been targeted by hackers.
3. US Secret Service and FBI investigation: An investigation into a network operating in Britain that was involved in the hacking of victims' social media accounts.
4. China's chipmakers affected by lockdowns: The shutdown of major business operations in Xi'an due to COVID-19 outbreaks has affected companies like Samsung and Micron, which rely heavily on Chinese suppliers.

Note that this is not an exhaustive list, but rather a selection of incidents mentioned in the provided context.

5.

Querying Ollama...without context: why is there a shortage of truckers ?

-- Result 1 ---

Document Metadata: {'Headline': 'Ford will start shipping Explorers without all the parts -- and add them later - CNN'}

Content: .Why car prices remain so stubbornly highFord is not the only automaker shipping vehicles without all the features they were designed to include. Last year General Motors was forced to temporarily sto ...

--- Result 2 ---

Document Metadata: {'Headline': "There's a shortage of truckers, but TuSimple thinks it has a solution: no driver needed - CNN"}

Content: . These are different skills and different systems altogether," he said. Removing the human element on long trucking routes has other benefits. The company says its trucks react 15 times faster than h ...

--- Result 3 ---

Document Metadata: {'Headline': "There's a shortage of truckers, but TuSimple thinks it has a solution: no driver needed - CNN"}

Content: (CNN)Right now, there's a shortage of truck drivers in the US and worldwide, exacerbated by the e-commerce boom brought on by the pandemic. One solution to the problem is autonomous trucks, and severa ...

--- Result 4 ---

Document Metadata: {'Headline': 'Opinion: Why inflation is about to peak - CNN'}

Content: . That could make shopping for your summer wardrobe, buying a new grill and loading up on hamburger buns easier to stomach as the weather improves. With fewer people getting sick, there should be more ...

--- Result 5 ---

Document Metadata: {'Headline': "Opinion: How we can solve the nation's affordable housing crisis - CNN"}

Content: . The big reason the US and UK are facing such a massive labor shortageLenders, the Federal Housing Administration and Fannie Mae and Freddie Mac can improve terms of financing for purchase/rehab loan ...

... LLM Response:

There are several reasons that contribute to the shortage of truckers in the United States. Here are some of the key factors:

1. ****Age and retirement****: Many experienced truckers are nearing or have retired, which has led to a shortage of skilled drivers. The Bureau of Transportation Statistics (BTS) estimates that there will be over 10% fewer qualified CDL holders by 2025.
2. ****Newer driver training programs****: Some trucking companies are shifting their focus from on-the-job training to online courses and automated systems, which can lead to a shortage of skilled drivers who require hands-on experience.
3. ****Increased competition****: The rise of alternative modes of transportation, such as ride-sharing services and electric vehicles, has increased competition for truckers. This competition is driving up wages and benefits, making it more challenging for truckers to afford retirement and support their families.
4. ****Changes in the industry****: The trucking industry has undergone significant changes in recent years, including consolidation, outsourcing, and the increasing use of technology. These changes can lead to a shortage of skilled drivers who are familiar with these new systems.
5. ****Driver shortages in other industries****: Many states have experienced driver shortages in other industries, such as aviation and railroading, which has led trucking companies to look for talent in these areas.
6. ****Lower pay rates****: Trucking companies often offer lower pay rates compared to other industries with similar levels of responsibility. This can make it difficult for truckers to afford retirement and support their families at a decent standard of living.
7. ****Increased emphasis on safety****: The rise of regulatory requirements, such as the Hours of Service (HOS) rule, has led to increased costs and challenges for trucking companies. This has driven up pay rates, making it more difficult for truckers to afford retirement.
8. ****Decline in truck stop infrastructure****: The decline of truck stops and rest areas has reduced opportunities for truckers to refuel, repair their vehicles, and socialize with colleagues, all of which are essential aspects of the job.
9. ****Limited incentives for new drivers****: Some trucking companies have limited or no incentives for hiring new drivers, such as bonuses or tuition reimbursement programs. This can make it more challenging for potential applicants to enter the industry.
10. ****Regulatory hurdles****: The rise of self-driving trucks and other autonomous vehicles has created regulatory challenges that may discourage some truckers from transitioning to these jobs.

These factors have contributed to a shortage of truckers in the United States, with many companies struggling to find qualified drivers who can meet their needs.

Querying Ollama...

Based on the following context, provide a response to the query: " {query} ".

Context: {context}

... LLM Response:

Based on the context provided, it appears that there is a shortage of truckers due to several factors. Here's why:

1. ****The pandemic has led to job displacement****: The COVID-19 pandemic has resulted in a significant number of workers losing their jobs, particularly those involved in transportation and logistics.
2. ****The e-commerce boom****: The rapid growth of e-commerce during the pandemic has created new demands on the transportation sector, leading to an increase in trucking-related work.
3. ****Lack of skilled drivers****: Many trucks require specialized skills, such as navigating long routes, dealing with hazardous conditions, and handling heavy loads. As a result, some truckers may be leaving the industry due to age or physical limitations.

In response to this shortage, several solutions are being explored, including:

1. ****Autonomous trucks****: Companies like TuSimple are developing autonomous trucks that can operate independently without human intervention.
2. ****Increased demand for trucking services****: As more people shop online and seek delivery services, the need for reliable truck drivers is likely to increase.

While these efforts aim to address the shortage of truckers, it's essential to acknowledge that this issue is complex and multifaceted. Addressing labor shortages in general requires a

comprehensive approach that includes training programs, improving working conditions, and encouraging workers to remain in the industry.